Quantification: 543-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2027.92	2066	537.141	px^2
major_axis_length_px	lacuna shape	81.5016	86.4661	16.9465	px
minor_axis_length_px	lacuna shape	33.8846	32.3799	6.9212	px
aspect_ratio	lacuna shape	2.4727	2.3986	0.6666	unitless
eccentricity	lacuna shape	0.8942	0.9087	0.0558	unitless
solidity	lacuna shape	0.8887	0.9115	0.0668	unitless
roots_count	headline	5.9167	5	3.7285	count
ring_length_r30_px	headline	271.583	256.5	101.779	px
ring_length_r60_px	reported	855.083	823	211.833	px
owned_length_px	ownership-dependent	2755.23	2967.43	1834.24	px
edge_count	ownership-dependent	128.417	136	82.4659	count
mean_edge_length_px	ownership-dependent	19.5274	21.8175	6.3195	px
perimeter_px	normalised	222.101	229.693	45.067	px
ring_area_r30_px2	normalised	8742.17	8905.5	1127.78	px^2
ring_area_r60_px2	normalised	22189.9	23189	2825.36	px^2
in_frame_fraction_r30	normalised	0.9965	1	0.0123	unitless
in_frame_fraction_r60	normalised	0.9814	1	0.0596	unitless
ring_density_r30	normalised	0.0308937	0.0328218	0.0101645	px^-1
ring_density_r60	normalised	0.0387431	0.0384869	0.00865819	px^-1
roots_per_100px_perimeter	normalised	2.63	2.5869	1.5678	per 100 px
ring_attached_length_r30_px	network v2	220.333	196	114.934	px
ring_attached_length_r60_px	network v2	638.667	645	307.191	px
ring_length_w_r30_px	network v2	305.526	282.894	116.118	px
ring_length_w_r60_px	network v2	976.6	946.936	245.859	px
sholl_crossings_r10	network v2	7	6	3.3303	count
sholl_crossings_r20	network v2	9.9167	9.5	3.5537	count
sholl_crossings_r30	network v2	12.25	11.5	2.6328	count
ring_width_mean_r30_px	width	7.3971	7.4123	0.428	px

Per field
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measure	value	units
analysed_area_px2	1.02424e+06	px^2
total_skeleton_length_px	37700	px
canalicular_length_density_per_px	0.0368077	px^-1
junction_count	923	count
junction_density_per_px2	0.00090116	px^-2
lacuna_count	12	count
lacunae_per_px2	1.172e-05	px^-2
skeleton_component_count	170	count

mean_component_length_px	221.765	px
median_component_length_px	19.5	px
field_length_density_w_per_px	0.0419148	px^-1
field_density_without_flagged_per_px	0.0368077	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	465	px
unattached_component_count	26	count
canalicular_length_density_attached_per_px	0.0368077	px^-1
field_length_density_with_unattached_per_px	0.0372617	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.9055	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	37700	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2494	90.6112	36.7713	2.4642	0.914	0.9096	0.4535	88.1403	271.181	False
2	3004	86.0791	52.8366	1.6292	0.7895	0.772	0.3181	207.791	919.033	False
3	1698	63.4601	34.3535	1.8473	0.8408	0.955	0.5365	257.425	340.805	False
4	1631	87.5017	30.956	2.8266	0.9353	0.741	0.0248	297.032	38.7695	False
5	1204	55.3793	28.6478	1.9331	0.8558	0.9177	0.1008	394.263	364.613	False
6	2066	81.299	34.8468	2.333	0.9035	0.8619	0.1545	472.247	745.532	False
7	1233	57.0332	27.8479	2.048	0.8727	0.9278	0.052	459.387	484.83	False
8	2066	100.448	26.9826	3.7227	0.9632	0.9018	0.236	550.262	877.61	False
9	2656	108.838	32.5449	3.3442	0.9542	0.8877	-0.0333	642.811	193.327	False
10	2113	91.9731	30.2288	3.0426	0.9444	0.9135	0.0369	728.295	633.991	False
11	2173	86.8531	32.2149	2.6961	0.9287	0.9358	0.0768	787.166	480.067	False
12	1997	68.5432	38.3841	1.7857	0.8285	0.9402	0.1037	859.482	288.042	False

Network per lacuna

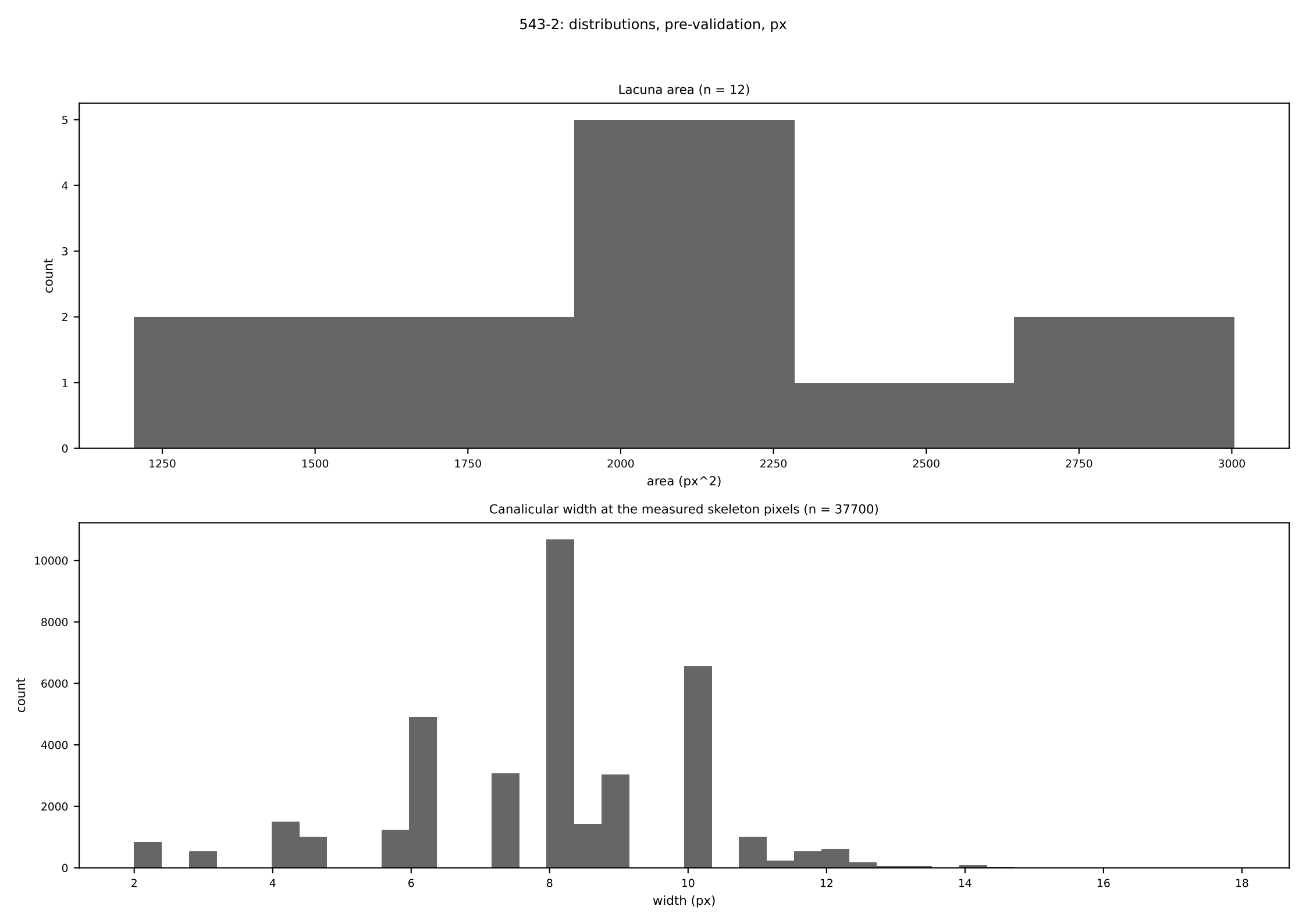
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	12	409	1290	4651.44	207	22.4707	7.7397
2	5	449	1161	3119.45	161	19.3754	7.2445
3	5	288	859	905.938	41	22.0961	7.4857
4	11	349	675	1174.52	65	18.0695	6.9562
5	4	255	827	4342.09	196	22.1535	7.0853
6	10	332	1001	5093.81	225	22.6391	7.6495
7	0	107	707	0	0	0	7.3907
8	1	131	517	339.593	15	22.6395	7.4614
9	6	232	819	2911.74	135	21.5684	7.4338
10	8	211	736	3023.12	137	22.0666	8.398
11	4	258	791	5141.74	240	21.4239	7.2076
12	5	238	878	2359.3	119	19.8261	6.7133

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	244.492	9510	23791	1	0.9843	0.0430074	0.0542222	4.9081
2	295.12	10519	26271	1	1	0.0426847	0.0441932	1.6942
3	172.894	7548	18977	1	1	0.0381558	0.0452653	2.8919
4	232.551	8756	18925	0.9575	0.7928	0.0398584	0.0356671	4.7301
5	154.267	7010	17818	1	1	0.0363766	0.0464137	2.5929
6	231.764	9080	22988	1	1	0.0365639	0.0435445	4.3147
7	155.924	7017	19168	1	1	0.0152487	0.0368844	0
8	254.25	9429	23720	1	1	0.0138933	0.021796	0.3933
9	277.078	10094	25807	1	1	0.0229839	0.0317356	2.1655
10	227.622	8939	23523	1	1	0.0236044	0.0312885	3.5146
11	225.522	8872	23390	1	1	0.0290802	0.0338179	1.7737
12	193.723	8132	21901	1	1	0.0292671	0.0400895	2.581

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	346	1129	473.475	1493.98	10	16	16
2	413	1056	499.191	1311.25	12	16	15
3	234	686	324.907	988.335	10	12	14
4	335	620	396.392	758.813	11	11	9
5	188	574	279.823	956.678	6	7	12
6	315	979	379.877	1148.93	9	11	11
7	24	98	119.326	803.926	1	7	9
8	69	287	151.782	596.286	4	5	9
9	163	413	253.924	937.193	5	9	11
10	190	670	235.924	829.47	5	6	11
11	202	688	285.966	883.169	6	10	15
12	165	464	265.723	1011.16	5	9	15


543-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1250
1500
1750
2000
2250
2500
2750
3000
Canalicular width at the measured skeleton pixels (n = 37700)
count
width (px)
0
2000
4000
6000
8000
10000
2
4
6
8
10
12
14
16
18